Background suppression in 3D affordance prediction

T14 3D affordance prediction

OpenAD on partial point clouds

CSE 59358-01 3D Machine Intelligence
Week 7 proposal / October 15, 2026

Task and research scope

Input

Object point cloud with 2,048 points and affordance text queries

Output

A score for each point and query, then a label from joint argmax

Scope

Localization on observed surfaces of partial point clouds

Baseline choice and setup

Public baseline

OpenAD PointNet++ with released pretrained weights

Data and evaluation

3D-AffordanceNet partial validation: 9,140 views
18 affordance queries plus the none class

Local execution

Apple M3 Pro, MPS, batch 1, seed 1
Frozen model with locally adapted evaluation

Measured baseline evidence

Archived OpenAD run	Input	mIoU
E01 historical baseline	Full shape	14.4121%
E01 historical baseline	Partial view	12.4964%
E05 all-view MPS evaluation	9,140 partial views	12.5364%

Metric: mean of 19 class IoUs, including none

Sampling and batch size differ across run records.
The planned comparison will use identical saved inputs.

Measured cost on one input

0.5456 s

Synchronized forward after setup
One cold run, no warm-up

1.295 GiB

Observed Metal-driver allocation high-water
5 ms sampling on Apple M3 Pro unified memory

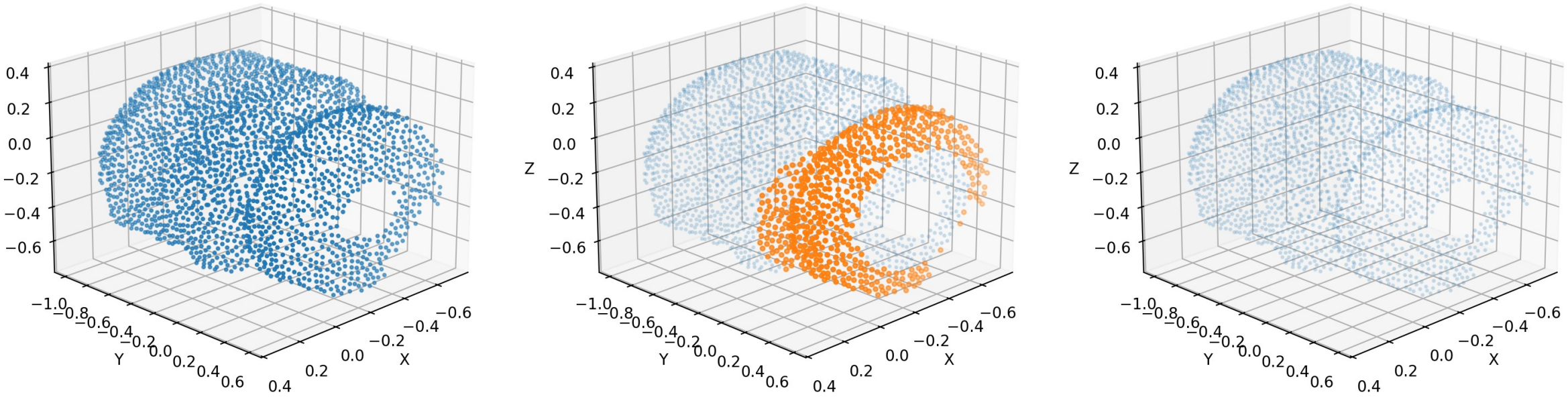
Observed failure: pour becomes none

Partial-view failure | pour | category: Vase | shape: edb8e199485be4e968747a7f12307bdb

Partial Point Cloud

Ground Truth: pour

OpenAD Prediction
IoU = 0.000



563 / 563 annotated pour points received none

Research question and hypothesis

Can adjusting only the none score improve localization on partial point clouds?

Hypothesis

A development-selected scalar bias increases 19-class mIoU with a background false-positive increase of at most 5 pp.

Support requires a positive paired 95% interval for the IoU gain and the predeclared false-positive guardrail.

One change, fixed comparison

Adjusted none log score = original log score - b

Comparison	Changed module	Controls
OpenAD, b = 0	None	Checkpoint, points, queries
OpenAD, frozen b	Output decision only	Same inputs and metric
Bias sweep / full-cloud control	One factor per analysis	Object IDs and view grouping

Select b on development objects. Lock it before validation.
Report 19-class mIoU, foreground IoU, recall, and background errors.

Milestones through Week 15

Weeks	Deliverable
7	Baseline manifest, failure figure, proposal
8-9	Object-grouped development split and bias calibration
10-11	Frozen partial-cloud comparison and object bootstrap
12-13	Full-cloud control and failure analysis
14-15	Reproducible report and final presentation

Main risk: recovered points receive the wrong affordance.
Fallback: a reproducible negative result with error trade-offs.

Proposed contribution

A controlled test of background suppression in a frozen 3D affordance model

Measured baseline. Observed failure. One scalar adjustment.
A result that can support or reject the hypothesis.

Appendix: reproduction record

OpenAD_PN2 / partial checkpoint

Repository commit
b082265ed085455a3fdc80b0d8d95dc51351e7b4

Checkpoint SHA-256
cbcfe9ff70448d97a7047f1ade817645a9be29dd2782fba57776792a8210f2ae

Python 3.10.21 / PyTorch 2.14.1 / seed 1 / batch 1
PointNet++ / 2,048 points for profiled sample / MPS

Local changes: model device handling, test runner, evaluator
Complete local diff and exact data manifest still need packaging.

Appendix: vocabulary and evaluation

Training query	Replacement evaluation query
grasp	grab
contain	accommodate
pourable	pour
wear	clothe
none	none

The current run tests replacement query names.
Separate seen/unseen aggregates require a verified grouping.

Appendix: evaluation safeguards

Split discipline

Development: official train objects, grouped by shape ID
Evaluation: official validation, already examined and exploratory

False-positive guardrail

Ground-truth none points predicted as any foreground class
Report the change against the unchanged baseline

Interpretation limits

The bias cannot improve ranking among foreground actions.
Scores do not establish real robot success.

Appendix: related work and scope

OpenAD / IROS 2023

Point-text alignment for affordance prediction
arXiv:2303.02401

Affogato / ECCV 2026

Large-scale data and Espresso grounding models
arXiv:2506.12009

Affostruction / CVPR 2026

Generative reconstruction and grounding from RGBD
arXiv:2601.09211